Question1 :

Observe the structure of the given **EMPLOYEE** table:

EMPNO	ENAME	SAL	DEPTNO	DNAME	LOC

Suggest if this table can be further normalized. Explain what normal form this table is in and how you can make it into the next normal form.

Solution:-

Step 1: Current Normal Form

The given table is in **First Normal Form (1NF)** because:

- All values are atomic (single-valued).
- There are no repeating groups or multivalued attributes.

However, there is a **partial dependency**:

- **DEPTNO → DNAME, LOC**
- Department name and location depend only on **DEPTNO**, not on the employee.

This causes data redundancy.

Step 2: Convert to Second Normal Form (2NF)

- The original table is in **1NF**.
- It can be further normalized into **2NF** by separating department information into a different table.

Separating employee details and department details into different tables.

EMPLOYEE Table

EMPNO	ENAME	SAL	DEPTNO
101	Rahul	35000	10
102	Amit	40000	20
103	Priya	38000	10

DEPARTMENT Table

DEPTNO	DNAME	LOC

10	HR	Delhi
20	Sales	Mumbai

SQL Queries

```
CREATE TABLE DEPARTMENT (
   DEPTNO NUMBER PRIMARY KEY,
   DNAME VARCHAR2(30),
   LOC VARCHAR2(30)
);

CREATE TABLE EMPLOYEE (
   EMPNO NUMBER PRIMARY KEY,
   ENAME VARCHAR2(30),
   SAL NUMBER,
   DEPTNO NUMBER,
   FOREIGN KEY (DEPTNO) REFERENCES DEPARTMENT(DEPTNO)
);
```

Sample Output

EMPLOYEE

EMPNO	ENAME	SAL	DEPTNO
101	Rahul	35000	10
102	Amit	40000	20
103	Priya	38000	10

DEPARTMENT

DEPTNO	DNAME	LOC
10	HR	Delhi
20	Sales	Mumbai

Question 2 :-

Observe the structure of the given **STUDENT** table:

ROLLNO	NAME	AGE	EXAM	MARKS	GRADE

Suggest if this table can be further normalized. Explain what normal form this table is in and how you can make it into the next normal form.

Solution:-

Step 1: Current Normal Form

The given table is in **First Normal Form (1NF)** because:

- All attributes contain atomic (single) values.
- There are no repeating groups.

However, the table has a **transitive dependency**:

- **MARKS → GRADE**
- Grade depends on Marks, not directly on the student.

This can lead to redundancy if the grading scheme changes.

Step 2: Convert to Third Normal Form (3NF)

- The given table is in 1NF.
- It can be normalized to 3NF by separating GRADE from the STUDENT table because GRADE depends on MARKS, not directly on the primary key (ROLLNO).

Separate the grade information into another table.

STUDENT Table

ROLLNO	NAME	AGE	EXAM	MARKS
101	Rahul	20	DBMS	85
102	Priya	21	OS	78
103	Amit	20	DBMS	92

GRADE Table

MARKS	GRADE
85	A
78	B+
92	A+

SQL Queries

```
CREATE TABLE STUDENT (
   ROLLNO NUMBER PRIMARY KEY,
   NAME VARCHAR2(30),
   AGE NUMBER,
   EXAM VARCHAR2(30),
   MARKS NUMBER
);

CREATE TABLE GRADE (
   MARKS NUMBER PRIMARY KEY,
   GRADE VARCHAR2(5)
);
```

Sample Output

STUDENT

ROLLNO	NAME	AGE	EXAM	MARKS
101	Rahul	20	DBMS	85
102	Priya	21	OS	78
103	Amit	20	DBMS	92

GRADE

MARKS	GRADE
85	A
78	B+
92	A+

Question 3 :

Observe the structure of the given **EMPLOYEE** table and suggest what kind of problem this table has and how to solve the problem.

Table Structure:

EMPNO	PROJECT_NO	NO_OF_DAYS	CUSTOMERNAME

Note: EMPNO and PROJECT_NO together form a **composite primary key**.

Solution:-

Step 1: Identify the Problem

The table is in **Second Normal Form (2NF)** because all non-key attributes depend on the composite key.

However, it has a **transitive dependency**:

- **PROJECT_NO → CUSTOMERNAME**
- CUSTOMERNAME depends only on PROJECT_NO, not on the entire composite key (EMPNO, PROJECT_NO).

This leads to:

- Data redundancy
- Update anomaly
- Insertion anomaly
- Deletion anomaly

Step 2: Convert to Third Normal Form (3NF)

Split the table into two tables.

EMPLOYEE_PROJECT Table

EMPNO	PROJECT_NO	NO_OF_DAYS
101	P01	30
101	P02	20
102	P01	25

PROJECT Table

PROJECT_NO	CUSTOMERNAME
P01	ABC Ltd.
P02	XYZ Pvt. Ltd.

SQL Queries

```
CREATE TABLE PROJECT (
  PROJECT_NO VARCHAR2(10) PRIMARY KEY,
  CUSTOMERNAME VARCHAR2(50)
);
```

```
CREATE TABLE EMPLOYEE_PROJECT (
  EMPNO NUMBER,
  PROJECT_NO VARCHAR2(10),
  NO_OF_DAYS NUMBER,
  PRIMARY KEY (EMPNO, PROJECT_NO),
  FOREIGN KEY (PROJECT_NO) REFERENCES PROJECT(PROJECT_NO)
);
```

Sample Output

EMPLOYEE_PROJECT TABLE

EMPNO	PROJECT_NO	NO_OF_DAYS
101	P01	30
101	P02	20
102	P01	25

PROJECT TABLE

PROJECT_NO	CUSTOMERNAME
P01	ABC Ltd.
P02	XYZ Pvt. Ltd.